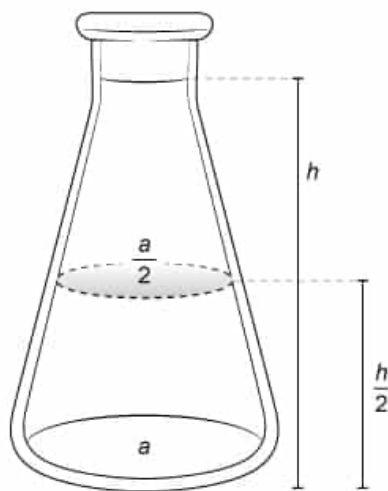




A flask with height h and base area A is filled with water. If the pressure due to the weight of the water is 2 atm halfway between the base and the top, what is the pressure due to the weight of the water at the base of the flask?

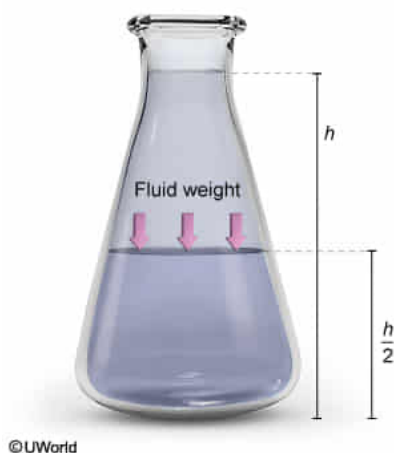
- ☐ A. 1,500 mm Hg [17%]
- ✓ ☒ B. 3,000 mm Hg [59%]
- ☐ C. 4,600 mm Hg [17%]
- ☐ D. 6,000 mm Hg [6%]

Correct

59% answered correctly

Explanation:

Hydrostatic pressure (P)



Hydrostatic pressure (P) describes the pressure fluid molecules in a static **column of fluid** exert on each other and their surroundings. In contrast to **osmotic pressure (Π)**, which causes fluid to migrate from regions of lower Π to regions of higher Π , P causes fluid to move away from areas of higher P toward areas of lower P .

P results from the **weight of the fluid** above a point within the fluid column. P is typically measured relative to atmospheric pressure at sea level, also called gauge pressure. The weight of the fluid is determined by the density of the fluid (ρ), the acceleration gravity ($g = 10 \text{ m/s}^2$), and the height (h) of fluid above the point:

$$P = \rho gh$$

As a result, container shape *does not* influence the hydrostatic pressure in a column of a noncompressible fluid with constant density.

In this question, a flask with variable cross-sectional area is filled with water. Because the water is not flowing, it forms a fluid column that generates hydrostatic pressure. The hydrostatic pressure at half the depth of the total flask height is 2 atm, which is twice the atmospheric pressure experienced by objects on Earth's surface at sea level:

$$P = \rho g \left(\frac{h}{2} \right) = 2 \text{ atm}$$

Therefore, the hydrostatic pressure experienced at the bottom of the flask is:

$$P = \rho gh = 4 \text{ atm}$$

Converting the units from atmospheres (atm) to millimeters of mercury (mm Hg), which is a unit derived from the height of a [mercury barometer](#) at atmospheric pressure, gives:

$$P = 4 \text{ atm} \cdot \frac{760 \text{ mm Hg}}{1 \text{ atm}} \approx 3,000 \text{ mm Hg}$$

Therefore, the pressure at the bottom of the flask is about 3,000 mm Hg, regardless of the flask widening at the bottom **(Choice B)**.

Educational objective:

Hydrostatic pressure describes the pressure caused by the weight of static fluid molecules within a column of fluid. The magnitude of hydrostatic pressure depends on the density of the fluid, the acceleration of gravity, and the height of the fluid; it is unrelated to container shape. Atmospheric pressure is 760 mm Hg at sea level.

Time spent: 0 sec
Subject:
 Physics

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System:
 Fluids